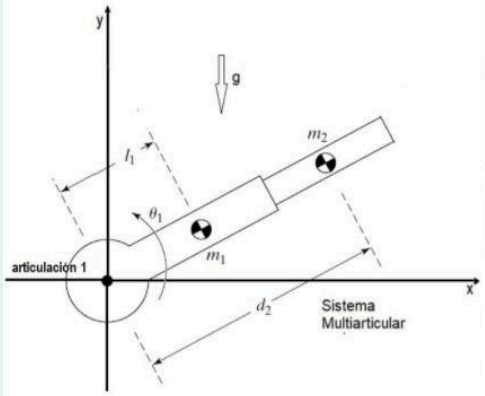
Tp3

Modelo dinámico RT. Determinar las expresiones dinámicas para la articulación 1 y articulación 2
Torque (T1) Art. 1= Expresión 1 Torque (T2) Art. 2=Expresión 5



1. $\tau_1 = (m_1 l_1^2 + I_{zz1} + I_{zz2} + m_2 d_2^2)\ddot{\theta}_1 + 2m_2 d_2 \dot{\theta}_1 \dot{d}_2 + (m_1 l_1 + m_2 d_2) g \cos(\theta_1)$
2. $\tau_1 = m_1(d_1^2 + d_2)\ddot{\theta}_1 + m_2 d_2^2 \ddot{\theta}_1 + 2m_2 d_2 \dot{d}_2 \dot{\theta}_1 + g\cos(\theta_1)[m_1(d_1 + d_2\dot{\theta}_1) + m_2(d_2 + \dot{d}_2)]$
3. $\tau_1 = (m_1 l_1^2 + I_{zz1} + m_2 d_2^2)\ddot{\theta}_1 + 2m_2 d_2 \dot{\theta}_1 \dot{d}_2 + (m_1 l_1 + m_2 d_2)\, g \operatorname{sen}(\theta_1)$
4. $\tau_2 = m_2 \ddot{d}_2 - m_2 d_2 \dot{\theta}_1^2 + m_2 g \operatorname{sen}(\theta_1)$
5. $\tau_2 = m_2 \ddot{d}_2 - m_2 d_2 \dot{\theta}_1^2 + m_2 g \cos(\theta_1)$
6. $\tau_2 = m_2 \ddot{d}_2 + m_1 \dot{d}_2 \ddot{\theta}_1 - m_2 d_2 \dot{\theta}_1^2 + m_2 g \operatorname{sen}(\theta_1)$

Sabiendo:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \tau$$

$$\mathcal{L} = k - u$$

$$\frac{d}{dt}\frac{\partial k}{\partial \dot{\Theta}} - \frac{\partial k}{\partial \Theta} + \frac{\partial u}{\partial \Theta} = \tau$$

$$\mathcal{L}(\Theta, \dot{\Theta}) = k(\Theta, \dot{\Theta}) - u(\Theta)$$

Busco E. Cinética de cada eslabón:

$$k_1 = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} I_{zz1} \dot{\theta}_1^2$$

$$k_2 = \frac{1}{2} m_2 (d_2^2 \dot{\theta}_1^2 + \dot{d}_2^2) + \frac{1}{2} I_{zz2} \dot{\theta}_1^2$$

$$k(\Theta, \dot{\Theta}) = \frac{1}{2}(m_1 l_1^2 + I_{zz1} + I_{zz2} + m_2 d_2^2)\dot{\theta}_1^2 + \frac{1}{2} m_2 \dot{d}_2^2$$

Busco E. Potencial de cada eslabón

$$u_1 = m_1 l_1 g \operatorname{sen}(\theta_1) + m_1 l_1 g$$

$$u_2 = m_2 g d_2 \operatorname{sen}(\theta_1) + m_2 g d_{2máx}$$

$$u(\Theta) = g(m_1 l_1 + m_2 d_2)\operatorname{sen}(\theta_1) + m_1 l_1 g + m_2 g d_{2máx}$$

Derivo:

$$\frac{\partial k}{\partial \dot{\Theta}} = \begin{bmatrix} (m_1 l_1^2 + I_{zz1} + I_{zz2} + m_2 d_2^2)\dot{\theta}_1 \\ m_2 \dot{d}_2 \end{bmatrix}$$

$$\frac{\partial k}{\partial \Theta} = \begin{bmatrix} 0 \\ m_2 d_2 \dot{\theta}_1^2 \end{bmatrix},$$

$$\frac{\partial u}{\partial \Theta} = \begin{bmatrix} g(m_1 l_1 + m_2 d_2)\cos(\theta_1) \\ g m_2 \operatorname{sen}(\theta_1) \end{bmatrix}.$$

$$\frac{d}{dt}\frac{\partial k}{\partial \dot{\Theta}} - \frac{\partial k}{\partial \Theta} + \frac{\partial u}{\partial \Theta} = \tau$$

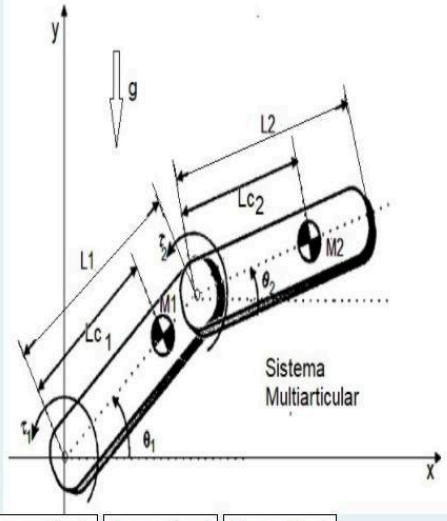
Aplico en

$$\tau_1 = (m_1 l_1^2 + I_{zz1} + I_{zz2} + m_2 d_2^2)\ddot{\theta}_1 + 2m_2 d_2 \dot{\theta}_1 \dot{d}_2 + (m_1 l_1 + m_2 d_2) g \cos(\theta_1),$$

$$\tau_2 = m_2 \ddot{d}_2 - m_2 d_2 \dot{\theta}_1^2 + m_2 g \operatorname{sen}(\theta_1).$$

DETERMINAR LAS EXPRESIONES DINÁMICAS PARA LA ARTICULACIÓN 1.

E. Cinética (K1) Art.1= **E. Potencial (u1) Art.1=**



1. $K1= \frac{1}{2} M1.Lc_1^2.\dot{\theta}_1^2 + I1.\dot{\theta}_1^2$
2. $U1= M1.g.Lc_1.\sin(\theta_1)$
3. $U1= M1.g.Lc_1.\cos(\theta_1)$
4. $K1= \frac{1}{2}(M1.L1^2.\dot{\theta}_1^2 + I1.\dot{\theta}_1^2)$
5. $U1= M1.g.Lc_1^2.\sin(\theta_1)$
6. $K1= \frac{1}{2}(M1.Lc_1^2.\dot{\theta}_1^2 + I1.\dot{\theta}_1^2)$

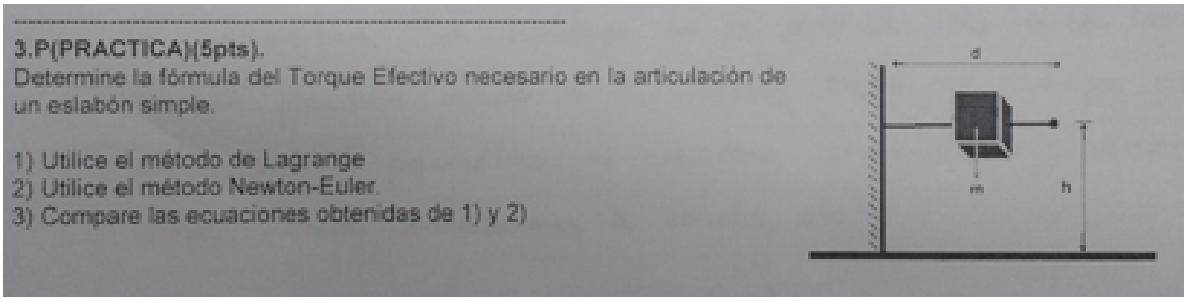
Expresión 1 | Expresión 4 | Expresión 6

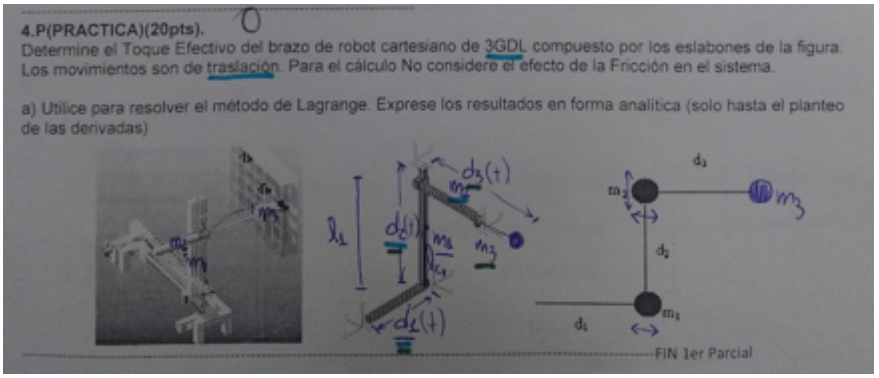
Expresión 2 | Expresión 5 | Expresión 3

K1= Expresión 6

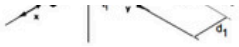
U1= Expresión 2

Esta en el power del profe, tene cuidado con la definicion de los angulos





a) Resuelva por el método de Lagrange



Las energías cinética y potencial para la articulación 1 seg

$$k_1 = \frac{1}{2} M_1 {v_1}^2 = \frac{1}{2} M_1 \dot{d_1}^2 \qquad (1)$$
$$u_1 = 0.$$

Para la segunda articulación,

$$k_2 = \frac{1}{2} M_2 {v_2}^2 \qquad (2)$$

siendo v_2

$$\begin{aligned} {v_2}^2 &= \dot{x_2}^2 + \dot{y_2}^2 \\ x_2 &= -d_1 \\ y_2 &= d_2. \end{aligned} \qquad (3)$$

Luego,

$$k_2 = \frac{1}{2} M_2 \left(\dot{d_1}^2 + \dot{d_2}^2 \right) \qquad (4)$$
$$u_2 = 0.$$

La función Lagrangiana resulta

$$\mathcal{L} = (k_1 + k_2) - (u_1 + u_2) = \frac{1}{2} \left(M_1 + M_2 \right) \dot{d_1}^2 + \frac{1}{2} M_2 \dot{d_2}^2. \qquad (5)$$

Calculamos las fuerzas en las articulaciones como

$$F_1 = \frac{d}{dt} \frac{\delta \mathcal{L}}{\delta \dot{d_1}} - \frac{\delta \mathcal{L}}{\delta d_1} \qquad (6)$$
$$\boxed{F_1 = \left(M_1 + M_2 \right) \ddot{d_1}}$$

$$F_2 = \frac{d}{dt} \frac{\delta \mathcal{L}}{\delta \dot{d_2}} - \frac{\delta \mathcal{L}}{\delta d_2} \qquad (7)$$
$$\boxed{F_2 = M_2 \ddot{d_2}.}$$